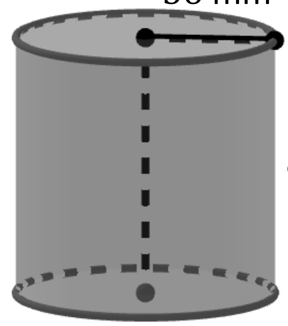
Grade 7 Name____________________
Unit 12 Assessment Date_________________
Calculator Period__________

1. A cylinder with dimensions in millimeters is shown.



a. Determine the surface area of the cylinder. Use $\pi = 3.14$. Round your answer to the nearest hundredth. Show your work. Be sure to include units in your answer.

Formula: $SA = 2B + Ph$ or $SA = 2\pi rh + 2\pi r^2$
work

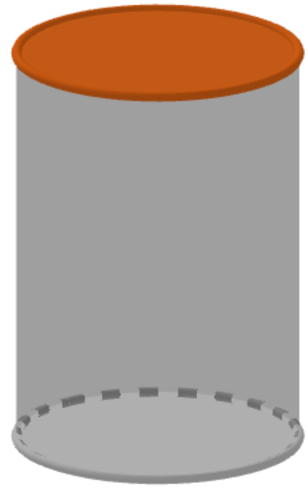
Surface Area	53,807.04 mm^2

b. Determine the volume of the cylinder. Use $\pi = 3.14$. Round your answer to the nearest tenth. Show your work. Be sure to include units in your answer.

Formula: $V = Bh$ or $V = \pi r^2 h$
work

Volume	955,162.9 mm^3

2. A glass cylindrical storage container has a diameter of 2.56 centimeters (cm) and a height of 12 cm. The container's top is made of bamboo.



a. Determine the surface area of the **glass part** of the cylindrical storage container. Round your answer to the nearest thousandth. Show your work. Be sure to include units in your answer.

Formula: $SA = 2B + Ph$ or $SA = 2\pi rh + 2\pi r^2$
work

Surface Area	101.657 cm^2 or 101.605 cm^2 or 101.698 cm^2

b. Determine the volume of the glass cylindrical storage container. Round your answer to the nearest hundredth. Show your work. Be sure to include units in your answer.

Formula: $V = Bh$ or $V = \pi r^2 h$
work

Volume	61.77 cm^3 or 61.74 cm^3 or 61.79 cm^3

3. A giant tube holds 120 lifesavers. Each lifesaver has a diameter of $2\frac{1}{2}$ inches (in.) and a height of $\frac{3}{20}$ in. What is the volume of the tube that contains the lifesavers? Round your answer to the nearest thousandth. Show your work. Be sure to include units in your answer.

Formula: $V = Bh$ or $V = \pi r^2 h$ work

Volume	88.357 in.^{3} or 88.313 in.^{3} or 88.393 in.^{3}

4. A cylindrical container of salt has a volume of 67.186 cubic inches (in^3). The diameter of the container is 3.4 inches (in.). What is the height of the container? Round your answer to the nearest hundredth. Show your work. Be sure to include units in your answer.

Formula: $V = Bh$ or $V = \pi r^2 h$ work

Height	7.40 in.